Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester	Category Code	Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits
Semester 1	HSC		Humanities - 1	2	0	0	2	2
	BSC		Mathematics – 1	3	1	0	4	4
	BSC		Physics	3	0	0	3	3
	BSC		Physics – Lab	0	0	2	2	1
	BSC		Environment Science	2	0	0	2	2
	ESC		Workshop Practices	0	0	2	2	1
	BSC		Biology for Engineers	2	0	0	2	2
	ESC		Introduction to Computer Programming – 1	1	0	0	1	1
	ESC		Introduction to Computer Programming – 1 Lab	0	0	2	2	1
	HSC		Universal Human Values	1	0	0	1	1
	HSC		Indian Knowledge System	2	0	0	2	2
				16	**1**	**5**	**23**	**20**
Semester 2	HSC		Humanities - 2	2	0	0	2	2
	BSC		Mathematics – 2	3	1	0	4	4
	BSC		Chemistry	3	0	0	3	3
	BSC		Chemistry - Lab	0	0	2	2	1
	ESC		Introduction to Electrical Engineering	3	0	0	3	3
	ESC		Introduction to Electrical Engineering-Lab	0	0	2	2	1
	ESC		Engineering Graphics	0	0	4	4	2
	HSC		Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC		Organizational Behaviour	1	0	0	1	1
	ESC		Introduction to Computer Programming – 2	1	0	0	1	1
	ESC		Introduction to Computer Programming – Lab	0	0	2	2	1
				13	**1**	**12**	**26**	**20**

<Course Code>					Mathematics – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs. / Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understand and apply principles of complex differentiation and integration effectively.
2. Solve systems, find eigenvalues, and apply transformations confidently.
3. Solve various ODEs, apply methods, and tackle engineering problems.
4. Analyze periodic functions, derive series, and apply in diverse applications.

UNIT 1 COMPLEX DIFFERENTIATION AND INTEGRATION **12 Hrs.**

Limit, Continuity, Differentiability of the function of a complex variable, Analytic function, Cauchy-Riemann equation (in Cartesian coordinates), Harmonic function and its significance, Singularities, Definition of a Complex line integral, Contour integrals, Cauchy integral theorem, Cauchy Integral formula (CIF), CIF for derivatives, Taylor's series and Laurent Series , Calculation of residues, Cauchy Residue theorem, Applications of residues to evaluate real definite integrals.

UNIT 2 MATRIX ALGEBRA AND ITS APPLICATIONS **10 Hrs.**

Solution of system of algebraic equation - Rank of a matrix, consistency of system of equation - Characteristic equation of a square matrix- Eigen values and Eigenvectors of a real matrix - Properties of Eigen values and Eigen vectors - Cayley-Hamilton theorem (without proof) - finding inverse of a matrix - Diagonalization of a matrix.

UNIT 3 ORDINARY DIFFERENTIAL EQUATIONS WITH APPLICATIONS **10 Hrs.**

Differential equations of Higher-order differential equations with constant coefficients, Rules for finding C.F. and P.I., Method of variation of parameters, Cauchy and Legendre's linear equations, Linear differential equations of second order with variable coefficients; Simultaneous linear equations with constant coefficients, applications of higher-order differential equations in solving engineering problems.

UNIT 4 FOURIER SERIES **10 Hrs.**

Periodic functions, Odd and even functions, Euler's formulae for Fourier series in an interval of length 2 pi, Change of interval, Dirichlet's conditions, Half range Sine and Cosine series, Complex Fourier series, Parseval's identity and its applications.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to

CO1 : Recall fundamental concepts of complex differentiation, matrix algebra, differential equations, and Fourier series.
CO2 : Understand the significance of complex differentiation, integration, Matrix, ODE and Fourier series in respective contexts.
CO3 : Apply the concept of complex function, Matrix, ODE and Fourier series to extract the solutions of engineering problems
CO4 : Analyze the use of complex variable Matrix, ODE and Fourier series in engineering problems.
CO5 : Assess the significance and effectiveness of mathematical concepts and theorems in solving real-world problems, particularly in engineering and scientific applications.
CO6 : Design and construct solutions to complex mathematical problems using a variety of techniques, including transformations, mappings, and advanced methods in differential equations.

TEXT/REFERENCE BOOKS

1. R.V. Churchill and J. W. Brown, Complex variables and Applications, McGraw-Hill.
2. J. M. Howie, Complex Analysis, Springer-Verlag.
3. R. K. Jain and S. R. K. Iyengar, Advanced Engineering Mathematics, Alpha Science.
4. Erwin Kreyszig, Advanced Engineering Mathematics", John Wiley.
5. G. Strang, Linear Algebra and its Applications, Cengage Learning.
6. K. Hoffman and R. A. Kunze, Linear Algebra", Prentice Hall of India.

<Course Code>					Engineering Chemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To develop the fundamental understanding about traditional materials.
2. To provide the knowledge about structural features, synthesis, properties of various categories of advanced materials.
3. To develop the knowledge and skills for different characterization techniques of materials.
4. To provide the knowledge about the role of chemistry in modern engineering applications.

UNIT I: CHEMISTRY OF ENGINEERING MATERIALS **12 Hrs.**

Traditional Materials: Introduction and classification of materials; metallic materials, polymeric, ceramic materials Advanced Materials: Introduction to nanomaterials: Properties and application; Carbonaceous materials (fullerene, carbon nanotube, graphene, etc.); Composite materials; Liquid crystals: Classification and Application

UNIT II: MODERN ANALYTICAL TECHNIQUES **10 Hrs.**

Instrumentation, principle and characterization of materials: X-ray diffraction (XRD), Electro analytical techniques; FTIR, UV-visible spectroscopy; Thermal analysis (TGA-DTA-DSC); Chromatographic techniques (GC, HPLC)

UNIT III: ADSORPTION, CATALYSIS AND KINETICS **10 Hrs.**

Adsorption - Characteristics, Classification, Application , Adsorption isotherms- Freundlich, Langmuir & BET Chemical Kinetics - Rate law, Arrhenius equation, Transition state theory, Collision theory; Complex reactions Catalysis - Homogeneous and Heterogeneous Catalysis; Mechanism of Catalysis; Industrial Applications of catalysts

UNIT IV: CHEMISTRY OF ENERGY DEVICES **10 Hrs.**

Principles of primary and secondary batteries, Fuel Cells and their operation principles, Principles and uses of supercapacitors; Photocatalytic hydrogen production: Principles and challenges; Traditional and new generation solar cells.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Remember the structural features and properties of different classes of traditional materials
CO2 : Classify advanced materials like nanomaterials, carbonaceous and composite materials, and liquid crystals
CO3 : Apply the skills by understanding various instrumental techniques for characterisation of materials.
CO4 : Analyse the key concepts in engineering chemistry viz. adsorption and chemical kinetics and laterally ponder over the applications of such concepts in engineering challenges.
CO5 : Justify the important insights into the industrial application of different types of catalysis via analysing mechanisms of catalysis.
CO6 : Develop the knowledge on the role of chemistry in various modern engineering applications such as in energy devices.

TEXT/REFERENCE BOOKS

1. W.D. Callister, An Introduction to Materials Science & Engineering, John Wiley & Sons.
2. MW Barsoum, Fundamental of Ceramics, IOP publishing.
3. T. Pradeep, Text book of Nanoscience and Nanotechnology, Mc. Graw Hill Education.
4. Murty, Shankar, B Raj, Rath, Murday, Textbook of Nanoscience and Nanotechnology", Springer.
5. V. Raghavan, Materials Science and Engineering, Prentice-Hall of India Private Limited.
6. A. Douglas, Donald Skoog, M.West, Principles of Instrumental Analysis, Cengage.
7. Jain & Jain, Engineering Chemistry, Dhanpat Rai Publishing Company.

<Course Code>					Engineering Chemistry Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To enhance and develop scientific and analytical skills
2. To relate concepts learned in chemistry and engineering to the real-world situations.
3. To acquire skills to perform laboratory experiments.
4. To demonstrate safe and proper use of standard chemistry glassware and equipment.

LIST OF EXPERIMENTS

1 **Iodometry**– To determine the strength of given copper sulphate solution by titrating against N/20 sodium thiosulphate (hypo) solution
2 **Iodimetry**– To determine the strength of given ascorbic acid by titrating against standard N/10 iodine solution
3 **Complexometric Titration**– To determine the total, permanent and temporary hardness of given water by complexometric titration using standard 0.01M EDTA solution
4 **pH metric titration**– To determine the strength of given HCl solution using a standard NaOH solution by performing a pH-metric titration
5 **Conductometric titration**– To determine the strength of given HCl solution using a standard NaOH solution by performing a conductometric titration
6 **Chemical Kinetics**– To study the kinetics of decomposition of sodium thiosulphate by a mineral acid
7 **Drawing chemical structures** - To Draw Chemical Structures of organic molecules using ChemDraw
8 **Colorimetric determination**: To determine the concentration of copper present in the effluent of electroplating industries by using colorimeter.
9 **Detection of biomolecule**: Detection of the presence of carbohydrates in test solution by using Benedict's reagent
10 **Preparation of drug molecule**: Preparation of Aspirin from salicylic acid
11 **Polymerization**– To prepare a polymer (Nylon 6,10), identify the functional groups by FT-IR

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall the concepts learned in chemistry and engineering to the real-world situations.
CO2 : Show the ability to identify, analyse and interpret the results from the experiments.
CO3 : Experiment with the instrumental method using conductometer and pH meter.
CO4 : Analyse compounds by titrimetric, gravimetric and instrumental methods.
CO5 : Determine the concentration of unknown solutions by spectrophotometric method.
CO6 : Predict the reaction rate and predict the order and rate constant.

TEXT/REFERENCE BOOKS

1. V. K. Ahluwalia, S Dhingra, A. Gulati, College Practical Chemistry, Universities Press.
2. J.B. Baruah, P Gogoi, Foundations of Experimental Chemistry, PharmaMed Press.
3. S. S. Sawhney, M. S. Jassal, S.P. Mittal, A Text Book of Chemistry Practical, Vol. I & II, APH Publishing Corp.

<Course Code>					Elements of Electrical and Electronics Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart knowledge on DC and AC circuits.
2. To learn construction, working principle and characteristics of transformer and induction machines.
3. To introduce students to various means for electrical safety, protection of electrical installations and Batteries.
4. To understand rectification through p-n junction diode, applications of diode and Transistor Characteristics.

UNIT I: DC CIRCUITS **10 Hrs.**

Electrical circuit elements (R, L and C), voltage and current sources, dependent and independent sources, Ohms Law, temperature co-efficient of resistance, Kirchhoff current and voltage laws, voltage and current divider circuit, Thevenin and Norton Theorems and their equivalents, maximum power transfer and superposition theorems, nodal and mesh analysis, star-delta transformation, Time domain analysis/natural response of first order RL and RC Circuit

UNIT II: AC CIRCUITS **10 Hrs.**

Generation of AC voltage, representation of sinusoidal waveforms, RMS values of different sinusoidal waveforms, Rectangular and Polar representation of phasor, Sinusoid representation in time and frequency domain, Analysis of single-phase ac series circuits consisting of R, L, C, RL, RC, RLC combinations, instantaneous power, average power and reactive power, complex power and power factor.AC parallel circuit and its solution in admittance form, resonance in AC series and parallel circuits, polyphase circuits, star and delta representation of polyphase circuits, power measurement in polyphase circuits.

UNIT III: INTRODUCTION TO ELECTRICAL MACHINES AND ELECTRICAL INSTALLATION **12 Hrs.**

Magnetic materials and its B-H characteristic, Faraday's Law of Electromagnetic Induction; **Single Phase Transformers:** Construction, working principle, types, EMF equation, ideal transformer, practical transformer, Operation of transformer on no-load and load, **Induction Machine:** construction, types of 3-phase induction motors, working principle, production of rotating magnetic field, operation, starting and running torques, Torque-slip characteristics, Power Stages in an induction motor.
Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing, Types of Batteries, Important Characteristics for Batteries. Elementary calculations for energy consumption, Electrical safety rules.

UNIT IV: BASIC OF ELECTRONICS AND ELECTRICAL INSTALLATION **10 Hrs.**

Semiconductor Diodes and Applications, Diode as a Switch, Diode as a Rectifier, Half Wave and Full Wave Rectifiers with and without Filters; Breakdown Mechanisms, Zener Diode – Operation and Applications; Opto-Electronic Devices–LEDs, Photo Diode and Applications; Silicon Controlled Rectifier (SCR) – Operation, Construction, Characteristics, Ratings, Applications.
Transistor Characteristics covering, Bipolar Junction Transistor (BJT) – Construction, Operation, Amplifying Action, Common Base, Common Emitter and Common Collector Configurations, Field Effect Transistor (FET) – Construction, Characteristics of Junction FET.

TOTAL HOURS 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Understand electrical circuits using network theorems.
CO2 – Compare the behaviour of R, L and C and their combinations in AC circuits.
CO3 – Analyze balanced polyphase systems in star and delta configurations.
CO4 – Understand the construction, working and basic characteristics of transformer and induction machines.
CO5 – Recognize the importance of protective devices, electrical safety measures and characteristics of Batteries.
CO6 -- Demonstrate application of different diodes in circuits and Understand the Transistor characteristics.

TEXT/REFERENCE BOOKS

1. J. Bird, "Electrical Circuit Theory and Technology", Routledge, Tailor and Francis Group, Sixth Edition.
2. D. P. Kothari and I. J. Nagrath, "Basic Electrical Engineering", Tata McGraw Hill.
3. B. L. Theraja, "Electrical Technology", Vol. 1, S. Chand Publication, New Delhi.
4. Surjit Singh, "Electrical Estimating and Costing", Dhanpat Rai and Co.
5. Boylestad and Nashlesky, "Electronic Devices and Circuit Theory", PHI.

					Elements of Electrical and Electronics Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To gain practical knowledge on DC and AC circuits
2. To learn operation of electrical instruments and electrical machines
3. To introduce students to various means for electrical safety, protection of electrical installations
4. To understand characteristics and applications of semiconductor diodes and transistors.

LIST OF EXPERIMENTS

1. Introduction to elements of electrical engineering laboratory and to study different electrical measuring instruments.
2. To Verify Ohm's law with linear resistors and find power dissipation in resistor.
3. To validate Thevenin and Norton theorem for DC circuits.
4. To validate Superposition and Maximum Power Transfer theorem for DC circuits.
5. To evaluate the AC R, L and R-L series circuit performance and to measure the active power, reactive power and apparent power connected with single phase AC supply.
6. To evaluate performance of AC R-L parallel circuit and to measure the active power, reactive power and the apparent power connected with single phase AC supply.
7. To perform a direct load test on a single-phase transformer and determine the efficiency and voltage regulation at different loads.
8. To measure three phase power using two wattmeter method.
9. To draw the time-current characteristics of MCB and study the function of ELCB.
10. To obtain the VI characteristic of silicon and germanium diodes.
11. To obtain reverse characteristics of Zener diode.
12. To study half wave, full wave and bridge rectifiers.
13. To determine the DC Characteristics of BJT in CE Configuration.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – To understand the basic operation of electrical equipment's & measuring instruments.

CO2 – To perform various network theorems for DC circuits.

CO3 – To understand the performance of AC circuit with R, L load.

CO4 – To evaluate star and delta configuration of polyphase system and measure power in polyphase system.

CO5 – To evaluate the performance of single phase transformer.

CO6 – To understand the working principle of semiconductor diodes, transistor characteristics and its applications.

<Course Code>					Engineering Graphics - Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	4	2	4	--	--	--	50	50	100

COURSE OBJECTIVES

1. To cover the fundamental of engineering drawing and standards used in drawing.
2. To explain the students to communicate ideas using orthographic and isometric projection methods.
3. To help students to use CAD software to prepare drawings.
4. To demonstrate the presentation of drawing using sketching and 3D modelling in CAD tool.

LIST OF EXPERIMENTS

1 Introduction to Engineering Graphics: Principles of engineering graphics and their significance, drawing instruments & accessories, lettering and numbering, types of lines, dimensioning methods, basic geometric drawing, reading a drawing.

2 Orthographic Projection: Introduction to projection, types of projection, 1st angle and 3rd angle projection

3 Isometric Projection: Principles of isometric projection – isometric scale, isometric views, conventions, conversion of isometric views to orthographic views and vice-versa

4 Projection of Solids and Development of Surface: Classification of solids, projections of solids like cylinder, cone, pyramid, and prism with its inclination to reference plane, development of surfaces of right regular solids - prism, pyramid, cylinder and cone.

5 Introduction of Computer Aided Engineering Drawing: Demonstrating knowledge of the theory of CAD software, use of software in drawing, CAD software user interface, commands, Coordinate System, menus and toolbars, planes, dimensioning, saving of files, Select and erase objects, zoom tools, and others

6 Basic sketching using CAD tool: Sketch entities using tools – origin, points, lines, circle, arcs, polygons, fillets and chamfer, trim, extend and offset, copy, cut, delete and others

7 Advanced sketching using CAD tool: Sketching entities using relation constrains, Mirror, Patterning, full definition of drawing and others

8 Basic 3D modelling using CAD tool: Extrude, cut, drawing on different planes, editing, symmetric, revolving, and others

9 Computer aided drawing sheets: Preparing drawing sheets, creating different views, section view, drawing templates, and others

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recalling the fundamentals of engineering graphics by considering basic rules of drawing, dimensioning, and labelling.
CO2 : Explain the principle of projection using orthographic and isometric projection.
CO3 : Represent the 2-dimensional drawing using CAD tool.
CO4 : Construct the 3-dimensional geometries using CAD tool.
CO5 : Apply the concept of engineering drawing by organizing drawing views and applying necessary dimensions by preparing drawing sheets
CO6 : Analyse the intricate details of solid using projection of solid, sectioning of solid and development of lateral surfaces.

TEXT/REFERENCE BOOKS

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing.
2. Shah P.J., Engineering Graphics, S. Chand Publishing.
3. Agrawal, B. & Agrawal C. M., Engineering Drawing, Tata McGraw Hill Publishers.
4. Hanifan R, Perfecting Engineering and Technical Drawing, Springer International Publishing Switzerland.
5. Corresponding Set of CAD Software Theory and User Manuals.

<Course Code>					Intoduction to Computer Programming II (For ME/CL/CH/BT)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

1. To develop a greater understanding of the foundational principles of the Python programming language
2. To acquaint with the data structures offered in Python.
3. To inculcate functional and logical problem-solving skills through programming.

UNIT 1: Fundamentals of Python Programming **4 Hrs.**

Introduction- History of python, Silent features, Advantage and Applications, Tools and its Framework, Variables,Input and Output

UNIT 2: Decision and Control Statements **4 Hrs.**

Operators in Python, Decision statements- If statements, If-else statements, Nested If-else statements, Control Statements - For, While, For-Else, While-Else, Continue and Break statements

UNIT 3: Data Structures and Collection **3 Hrs.**

String manipulation, List and slicing, Tuple, Set and Dictionary, Numpy and Dataframes

UNIT 4: Function and File Handling **3 Hrs.**

Built-in and user defined functions, Actual and Formal Parameters, Scope, Create, Read and Write text files

14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics of python programming such as syntax, datatypes, variable declarations.
CO2 - Understand the fundamental of data structures in python such as list, tuple, set, dictionary and strings.
CO3 - Understand the use of built-in functions, custom modules and packages.
CO4 - Apply Python's features to solve problems and implement decision-making and loops.
CO5 - Evaluate the effectiveness of functions and judge Python code quality.
CO6 - Evaluate Python scripts, create functions, and design programs for file handling tasks.

TEXT/REFERENCE BOOKS

1. John V Guttag, Introduction to Computation and Programming Using Python, PHI.
2. Allen Downey, Jeffrey Elkner and Chris Meyers, How to think like a Computer Scientist, Learning with Python, Green Tea Press.
3. Martin C. Brown, Python: The Complete Reference, Osborne, McGraw-Hill.
4. R. Nageswara Rao, Core Python Programming", Dreamtech Press.

<Course Code>					Introduction to Computer Programming II Laboratory (For ME/CL/CH/BT)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES

1. To develop a greater understanding of the foundational principles of the Python programming language
2. To acquaint with the data structures offered in Python.
3. To inculcate functional and logical problem-solving skills through programming.

LIST OF EXPERIMENT

1. **Installation:** Installing Python on Windows, Mac, and Linux.
2. **Introduction of Different Tools & Framework:** Introduce and utilizes different tools and frameworks for python program development
3. **Fundamentals of Python Programming**
   Variable declaration, initialization, different syntax and keywords, Input/output operations
4. **Decision Statements**: Operators- Arithmetic, Logical, etc., Decision-making - if, if-else, and nested if-else statements.
5. **Control Statements:** Control statements such as for loops, while loops, and their application, Continue and break statement
6. **Data Structure:** Demonstrate string manipulation techniques, including string concatenation, list, slicing, and formatting.
7. **Python Collection:** Implement programs that illustrate the characteristics and usage of tuples, sets, and dictionaries, numpy, dataframes.
8. **Function:** Develop Python functions to perform common tasks, including mathematical operations, string manipulation, and data processing, Built-in and user-defined functions, parameter actual and formal parameters, Implement functions with different scopes.
9. **File Handling:** Develop Python scripts that create, read, and write text files, showcasing file handling operations such as opening, reading, writing, and closing files.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Apply Python basics tools, laying the foundation for practical usage.
CO2 - Apply Python's features to manipulate data structures effectively for solving real-world problems.
CO3 - Apply decision-making and looping to solve practical challenges.
CO4 - Analyze Python code to select appropriate constructs and data structures for optimal performance.
CO5 - Evaluate function effectiveness in Python programs for code quality and efficiency.
CO6 - Develop Python scripts and programs with robust file handling, utilizing modular functions for enhanced organization and reusability.

TEXT/REFERENCE BOOKS

1. John V Guttag. Introduction to Computation and Programming Using Python, PHI.
2. Allen Downey, Jeffrey Elkner and Chris Meyers "How to think like a Computer Scientist, Learning with Python, Green Tea Press.
3. Martin C. Brown, Python: The Complete Reference, Osborne, McGraw-Hill.
4. R. Nageswara Rao, Core Python Programming, Dreamtech Press.